

Juncheng Zhu

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EDUCATION

University of Waterloo

Candidate for Honours Computer Science, Co-op

Waterloo, ON

Sept. 2025 – Present

GPA: 90% (Term 1A) | **Awards:** René Descartes National Scholarship (\$25k), President's Scholarship.

Relevant Coursework: Designing Functional Programs (90%), Calculus I (92%), Algebra (98%).

HONOURS & AWARDS

USACO Gold Division (Algorithm Design)

Euclid Contest: Honour Roll (95/100, Top 0.1%)

AMC 12: Honor Roll of Distinction (Top 1%)

Fermat Contest: National Champion (Full Marks)

CMO Qualifier (Top 50 in Canada)

IMMC (Modeling): National 2nd Place (2025)

CCC: Honour Roll (Top 2.5%)

AIME: Distinction (Score: 11/15)

TECHNICAL SKILLS

Languages: Python, C/C++ (STL), TypeScript, JavaScript (ES6+), Java, SQL, Racket

Frontend: React, Vite, HTML5 Canvas, Tailwind CSS, Chart.js, HTML/CSS

Backend/AI: PyTorch, Node.js, Flask, NumPy, Pandas, SQLite, SQLAlchemy, Reinforcement Learning

DevOps/Tools: Git, Docker, Linux/WSL, DigitalOcean, Vercel, Postman, LaTeX

PROJECTS

Sakura Rain Extension | *React, Vite, JS, Canvas, Value Noise* (Published on Web Store)

Jan. 2026

- Modernized legacy JS interface into a modular **React 18** and **Vite** architecture, significantly enhancing maintainability.
- Replaced standard animations with a custom **1D Value Noise** algorithm and power-curve physics, solving the "pendulum effect" to simulate stochastic, organic wind patterns.
- Engineered a high-performance rendering engine using **HTML5 Canvas** batch processing, maintaining **60 FPS with 1500+ procedural petals**.
- Developed a Chrome Extension popup UI with React state management for real-time physics tuning.

Deep Q-Learning Snake Agent | *Python, PyTorch, Reinforcement Learning*

Dec. 2025

- Built and trained an autonomous agent using **Deep Q-Networks (DQN)** to master the classic Snake game.
- Diagnosed "Model Collapse" issues; researched and implemented **Epsilon Decay** and **Experience Replay** to balance exploration vs. exploitation phases.
- Designed a custom **Manhattan Distance** reward function to solve the sparse reward problem, accelerating training convergence by over 40%.
- Analyzed failure cases (CNN vs. MLP) and documented hyperparameter experiments in a technical log.

PyFinance Tracker | *Python, Flask, SQLite, Chart.js, SQLAlchemy*

Oct. 2025 – Present

- Deployed a secure financial dashboard on **DigitalOcean**, maintaining **99.9% uptime** for continuous expense tracking.
- Implemented secure user authentication and session management using Flask-Login and password hashing.
- Designed a normalized SQLite schema and optimized SQLAlchemy queries, reducing query latency by **30%**.
- Integrated Chart.js to render dynamic financial reports capable of visualizing **1,000+ data points** without lag.

LEADERSHIP & EXPERIENCE

Math Club President

Bayview Secondary School

Richmond Hill, ON

Sept. 2024 – June 2025

- **Mentored** 30+ students in advanced problem-solving techniques for Euclid and CSMC contests.
- Organized weekly training sessions and mock contests, fostering a collaborative learning environment.
- Led the school team to achieve **13th Place** globally in the HMMT 2024 Team Guts Round.